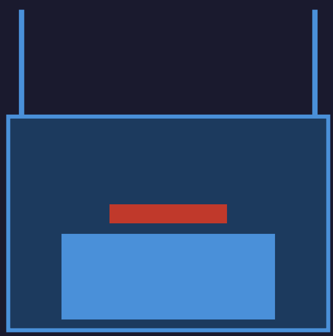
LOW COST 3D PRINTER

USER MANUAL



Designed by:

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FDM • Arduino Mega 2560 • RAMPS 1.4 • Marlin 2.1.2.5

1. Introduction

This user manual is intended to guide anyone who wants to operate the custom-built **Fused Deposition Modeling (FDM) 3D printer** designed and assembled by our team.

The goal was to create a **low-cost, fully functional 3D printer** that produces quality prints with PLA filament and is easy to maintain or upgrade.

This manual covers technical specifications, setup instructions, daily operation, maintenance, and troubleshooting.

2. Technical Specifications

Motion System	COREXYZ motion (Cartesian)
Build Volume	200 × 200 × 170 mm
Controller Board	Arduino Mega 2560 + RAMPS 1.4 Shield
Stepper Drivers	A4988
Stepper Motors	NEMA 17 – 1× X, 1× Y, 2× Z, 1× Extruder
Power Supply	12 V / 30 A (Industrial SMPS)
Firmware	Marlin 2.1.2.5
Printer Type	FDM – Fused Deposition Modelling
Hotend	E3D V6
Nozzle Sizes	0.4 mm (recommended), 0.6 mm, 0.8 mm
Heat Bed	RepRap 200×200 mm – 12 V / 10 A
Display	12864 Graphical LCD with rotary encoder

3. FDM Technology & Motion System

3.1 Fused Deposition Modeling (FDM)

FDM is one of the most accessible 3D printing technologies. Plastic filament is heated inside the hotend and extruded through a nozzle layer by layer to build up the object.

3.2 XYZ Cartesian Motion

Movement happens along three perpendicular axes:

- **X-axis** – left / right (extruder assembly, GT2 belt driven)
- **Y-axis** – front / back (print bed, GT2 belt driven)
- **Z-axis** – up / down (lead-screw driven, two synchronised motors)

4. Electrical Components

4.1 Arduino Mega 2560

- Main controller based on the ATmega2560 microcontroller.
- Interfaces with RAMPS 1.4 to control motion, temperature, and user inputs.

4.2 RAMPS 1.4 Shield

- Mounts directly on top of the Arduino Mega.
- Provides slots for stepper drivers and terminals for heater, thermistor, endstops, fans.

4.3 A4988 Stepper Motor Driver

- One driver per motor (X, Y, Z1, Z2, Extruder).
- Features adjustable current control and microstepping support.

4.4 NEMA 17 Stepper Motors

- High-precision positioning; rated 1.2 A – 1.8 A per phase.
- Drives X, Y, Z axes and the extruder.

4.5 Power Supply – 12 V / 30 A

- Industrial SMPS type with short-circuit protection and built-in cooling fan.

5. Axis Details, Hotend & Heat Bed

5.1 X-Axis

- Moves extruder assembly left/right via GT2 belt + 20-tooth pulley.
- 8 mm SS smooth rods + LM8UU linear bearings for smooth, stable motion.

5.2 Y-Axis

- Controls front-to-back movement of the print bed via GT2 belt.

5.3 Z-Axis

- Two NEMA 17 motors drive 8 mm lead screws through 5×8 mm flexible couplings.
- Vertical guide: 8 mm SS rods + LM8UU linear bearings.

5.4 Hotend – E3D V6

- **Heater block** – heats filament to target temperature.
- **Nozzle** – controls extrusion width (0.4 mm recommended).
- **Heat break** – prevents heat creep into the cold zone.
- **Heat sink** – cools the upper section of the hotend.
- Operating temperature range: **180 °C – 260 °C**.

5.5 Heat Bed – RepRap 200×200 mm

- PCB heat bed with PEI sheet surface for good first-layer adhesion.
- 12 V DC, up to 10 A; 100 kΩ NTC thermistor (EPCOS B57560G104F).
- Maximum recommended temperature: 110 °C.

6. Mechanical Parts

SS Rod	8 mm diameter smooth stainless steel
Lead Screw	8 mm, 2-start (Z-axis)
Linear Bearing	LM8UU – 8 mm
Flexible Coupling	5×8 mm (motor shaft to lead screw)
Lead Screw Nut	Brass T8 nut
Endstop	Mechanical micro-switch type
GT2 Pulley	20-tooth, 8 mm bore
GT2 Timing Belt	2 mm pitch, 6 mm wide

7. Display – 12864 LCD

Resolution	128 × 64 pixels (Graphical LCD)
Controller	ST7920
Interface	SPI
Input	Rotary encoder with push button
Features	SD card slot, buzzer, navigation menu
Voltage	5 V from controller board

8. Software

8.1 Marlin Firmware – v2.1.2.5

Marlin is an open-source firmware for 3D printers running on Arduino Mega 2560 + RAMPS 1.4.

MOTHERBOARD	BOARD_RAMPS_14_EFB
BAUDRATE	250000
AXIS STEPS/MM	X: 102 | Y: 103 | Z: 400 | E: 95
INVERT_X_DIR	true

INVERT_Y_DIR	false
INVERT_Z_DIR	true
MAX FEEDRATE (mm/s)	X/Y: 300 | Z: 5 | E: 25
MAX ACCELERATION	X/Y: 1000 | Z: 100 | E: 1000

Edit **Configuration.h** for basic settings. Edit **Configuration_adv.h** for advanced features.

8.2 Slicing Software – Ultimaker Cura / PrusaSlicer

Converts 3D models (.stl) into layer-by-layer G-code instructions for the printer.

Layer Height	0.2 mm (standard quality)
Wall Line Count	2 (0.8 mm total wall thickness)
Infill Density	20 % – Grid pattern
Print Speed	50 mm/s
Travel Speed	120 mm/s
Nozzle Temperature	200 °C (PLA)
Bed Temperature	60 °C (PLA)
Retraction	5 mm distance, 45 mm/s speed
Cooling Fan	Enabled – 100 %
Build Plate Adhesion	Brim – 5 mm width

8.3 Host Software – Pronterface (Printrun)

Connects to the printer over USB for direct G-code control, temperature monitoring, and calibration.

- Connect via USB – select the correct COM port.
- Manually jog X, Y, Z axes and extruder.
- Key G-codes: G28 (home all), M104 S200 (set nozzle), M140 S60 (set bed), M500 (save EEPROM).
- Monitor real-time temperatures of hotend and heated bed.
- Calibrate Z-offset, PID values, and steps/mm.

9. Step-by-Step Printing Instructions

Step 1 – Power On

Turn on the machine using the main switch on the power supply.

Step 2 – Preheat

Via the LCD or Pronterface: set nozzle to 200 °C and bed to 60 °C for PLA. Wait for both to reach target temperature before proceeding.

Step 3 – Load Filament

Feed the filament through the PTFE tube into the extruder. With the nozzle hot, push filament until it extrudes cleanly from the nozzle tip.

Step 4 – Insert SD Card / Connect USB

SD: insert card containing the .gcode file. USB: open Pronterface and connect to the correct COM port.

Step 5 – Level the Bed

Use the four corner knobs to set a uniform gap of ~0.1 mm (one sheet of paper) between the nozzle and bed at all four corners.

Step 6 – Start the Print

LCD: navigate Print from SD and select your .gcode file. Pronterface: Load File, then click Print.

Step 7 – Monitor First Layer

Watch the first layer carefully. Lines should be well-adhered and evenly spaced. Pause or abort if warping, poor adhesion, or stringing is observed.

Step 8 – Wait for Completion

Do not move the printer during operation. Check periodically for errors or failures.

Step 9 – Remove the Print

Let the bed cool to room temperature. Gently remove the object with a spatula or scraper.

Step 10 – Power Off

Power off the printer and clean the bed surface with IPA if needed.

10. Bed Levelling

Correct nozzle-to-bed distance is critical for first-layer adhesion:

Nozzle too high	Insufficient pressure on filament. Small contact area. Raft may detach mid-print.
Correct distance	Filament pressed into bed slightly. Maximises surface contact while maintaining flow.
Nozzle too low	Not enough clearance for extrusion. Can damage nozzle or bed surface.

11. Troubleshooting Guide

Issue	Possible Cause	Solution
First layer not sticking	Bed not leveled, cold bed, dirty surface	Re-level bed, increase bed temp, clean with IPA

Filament not extruding	Nozzle clogged, extruder gear loose, filament tangled	Heat nozzle and clean, tighten gear, check spool
Layer shifting	Loose belts, driver overheating, frame flex	Tighten belts, check driver cooling, reinforce frame
Warping at corners	Bed too cold, poor adhesion, strong cooling fan	Increase bed temp, use brim/glue, reduce fan early
Stringing / oozing	Retraction settings off, nozzle too hot	Adjust retraction distance/speed, lower temp
Under-extrusion	Partially clogged nozzle, incorrect flow rate	Clean nozzle, increase flow rate slightly
Over-extrusion	Flow rate too high, wrong filament diameter	Calibrate flow rate and filament diameter
Z-banding	Lead screw bent, inconsistent Z movement	Check/straighten Z screw, lubricate Z-axis
Nozzle dragging	Z-offset too low, over-extrusion	Re-adjust Z-offset, lower flow rate
Print stops mid-way	SD card issue, board overheating, power drop	Use quality SD card, improve cooling, use UPS

12. Safety Instructions

WARNING: *Always observe these precautions when operating the printer.*

- Use the printer in a **well-ventilated area** – PLA fumes can accumulate.
- **Do not touch** the hot nozzle or heated bed while printing or immediately after.
- Keep hands, hair, and tools **away from moving parts** during operation.
- **Turn off power** before performing any wiring or mechanical maintenance.
- Keep the printer **away from flammable materials** at all times.
- Wait for all parts to **cool down completely** before touching after a print.
- **Do not leave** the printer unattended during long print jobs.
- Keep **children and pets** away from the printer while it is operating.

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